The 6th Asia Exhibition of Innovations and Invention Hongkong

Prasarnmit Demonstration School (Secondary)
SRINAKHARINWIROT UNIVERSITY

TrueRehab-RF

AI-system for TrueRecovery utilizing mmWave radar

Background & Problem

Home-based physical rehabilitation help patients recover quickly and improve treatment outcomes, such as individuals recovering from stroke.

However, when patients become fatigued during rehabilitation, **they might use other muscle groups or adopt incorrect body postures.** These compensatory movements may lead to:

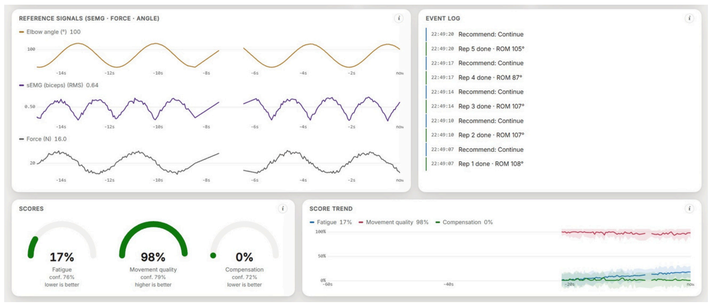
- Muscle Strain
- secondary Injuries
- slow Progress
- Ineffective treatment

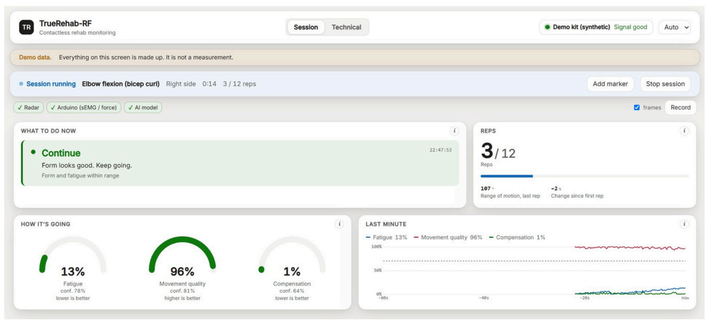
Innovation concept

TrueRehab-RF is a contactless rehabilitation monitoring system that **integrates artificial intelligence (AI) with millimeter-wave radar (mmWave radar) technology** to analyze patients' movements.



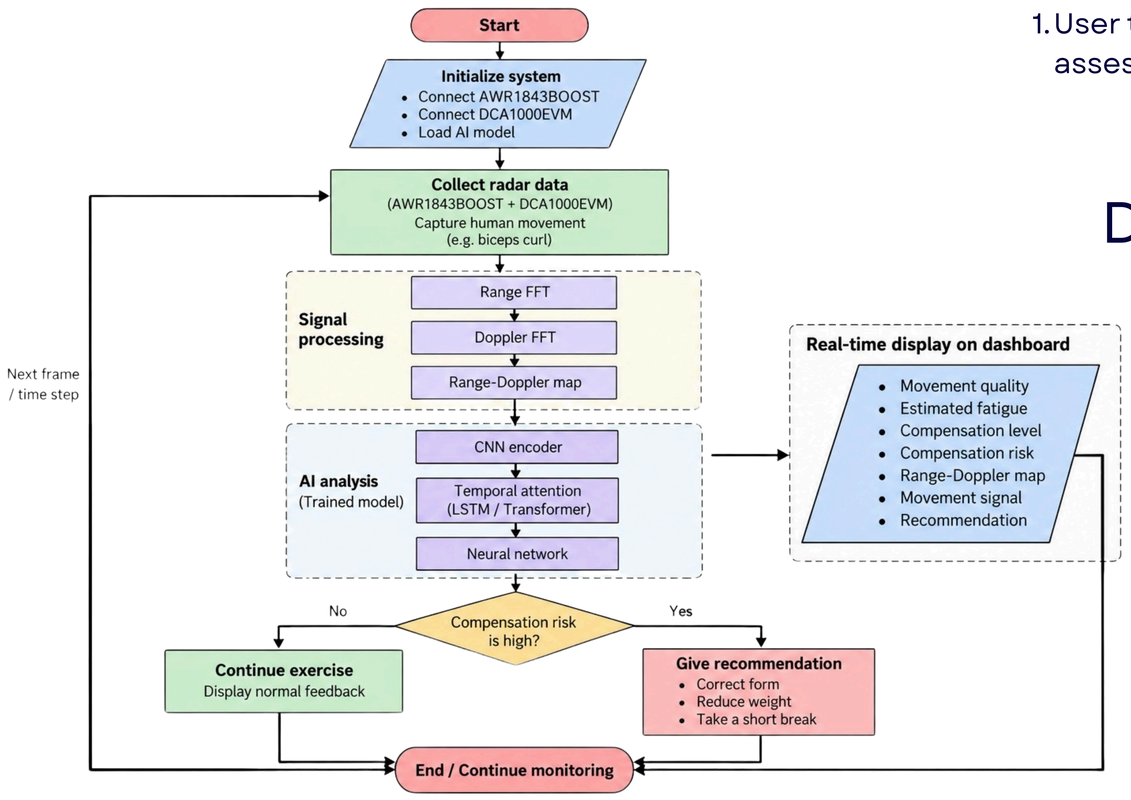
Dashboard





- **Improve rehabilitation effectiveness**
- **Support independent home rehabilitation**
- **Enhance patients' quality of life**
- **Detect incorrect muscle usage**
- **Reduce the risk of re-injury**

Operation Flowchart



Key development

1. User testing and performance assessment
2. Rehabilitation problem analysis
3. mmWave radar data collection
4. Movement and posture evaluation system
5. AI model development and training

Distinctive point

- **Privacy-friendly**
- **Contactless monitoring**
- **Real-time posture analysis**
- **Early detection of incorrect muscle usage**
- **Suitable for home rehabilitation**

Patients	Families & Caregivers	Hospitals	Physical Therapists
Individuals who require physical rehabilitation.	People who assist patients during rehabilitation.	Healthcare facilities that provide rehabilitation services.	Healthcare professionals who monitor patients' progress.

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